

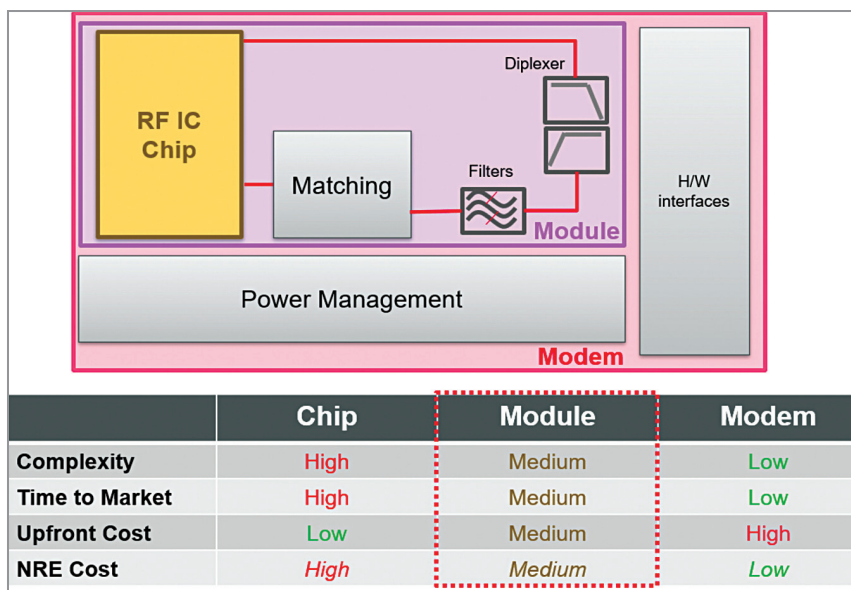


Fig. 2: TCO: Expensive certification, engineering cost (Lower upfront cost might be tempting, but TCO could be far higher until production volume hits few tens of thousands)

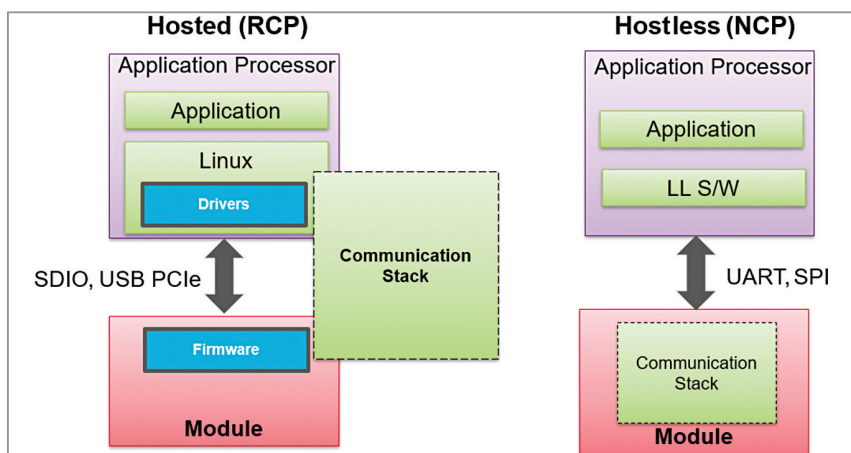


Fig. 3: Hostless vs host architecture: Chip unit dedicated for wireless vs chip unit dependent on MPU/MCU

distances, are helpful for remote IoT deployments and can run on a battery.

In contrast, short-range technologies like Wi-Fi and Bluetooth, often found in smartphones and consumer devices, deliver higher data rates but are limited in range and are more power-hungry, falling under what is termed a personal area network.

The wireless buffet: System-on-chip

The starting point for wireless product integration often begins with a system-on-chip (SoC), which

integrates the wireless transceiver, memory, and sometimes a GPU and NPU into a single block. SoCs can be derived from either microcontrollers or microprocessors and are particularly advantageous in space-constrained designs.

The key trade-off here is efficiency versus flexibility. SoCs offer

streamlined integration and power efficiency but limit customisation.

For applications demanding more flexibility, designers may turn to external wireless companion chips. These allow the MCU or MPU to be paired with a discrete wireless module. Some options are:

- RF ICs (radio frequency integrated circuits) offer raw radio functionality but require custom front-end design for matching, filtering, and antenna integration. RF ICs cannot be directly used on any PCB and therefore need proper front-end design.

- Modules simplify integration by embedding both the RF IC and the required front-end circuitry. These are ideal for mid-volume production, balancing design control with ease of use.

Host vs hostless architecture

When building wireless products, one of the key decisions is whether to go for a hosted or hostless architecture. It all comes down to where the wireless communication logic, or stack, sits in the system.

A hosted (radio co-processor) wireless chip handles low-level functions like sending signals, while the main processor manages higher-level networking. This offers more flexibility but demands a powerful host.

On the other hand, in the case of a hostless architecture, called networking co-processor, the wireless chip manages the entire communication stack independently. This is easier to integrate, especially in low-power or compact devices. It is a plug-and-play style solution, and it is ideal for teams with limited RF or firmware expertise.

Why 50 ohms?

The standardisation of 50-ohm impedance in RF systems is not arbitrary; it balances two performance factors: power handling and signal attenuation. While 30-ohm systems offer superior power handling and 77-ohm lines minimise signal loss, 50-ohm represents a practical compromise between the two, supporting a wide range of use cases without forcing designers to trade efficiency for signal strength. In short, 50 ohms is where electrical performance and design practicality converge.